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```
===== SMART CITY ROAD NETWORK =====
1. Add Junction
2. Add Road and Construction Cost
3. Display Junctions
4. Display Available Roads
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 1
Enter junction name: city centre
Junction added with ID: 0

===== SMART CITY ROAD NETWORK =====
1. Add Junction
2. Add Road and Construction Cost
3. Display Junctions
4. Display Available Roads
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 1
Enter junction name: north zone
Junction added with ID: 1

===== SMART CITY ROAD NETWORK =====
1. Add Junction
2. Add Road and Construction Cost
3. Display Junctions
4. Display Available Roads
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 1
Enter junction name: south zone
Junction added with ID: 2

===== SMART CITY ROAD NETWORK =====
1. Add Junction
2. Add Road and Construction Cost
3. Display Junctions
4. Display Available Roads
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 1
Enter junction name: east zone
Junction added with ID: 3
```

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```
===== SMART CITY ROAD NETWORK =====
1. Add Junction
2. Add Road and Construction Cost
3. Display Junctions
4. Display Available Roads
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 1
Enter junction name: west zone
Junction added with ID: 4

===== SMART CITY ROAD NETWORK =====
1. Add Junction
2. Add Road and Construction Cost
3. Display Junctions
4. Display Available Roads
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 2

--- Smart City Junctions ---
0 : city centre
1 : north zone
2 : south zone
3 : east zone
4 : west zone
Enter first junction ID: 0
Enter second junction ID: 1
Enter road construction cost: 4
Road added successfully!

===== SMART CITY ROAD NETWORK =====
1. Add Junction
2. Add Road and Construction Cost
3. Display Junctions
4. Display Available Roads
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 2

--- Smart City Junctions ---
0 : city centre
1 : north zone
2 : south zone
```

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--- Smart City Junctions ---
0 : city centre
1 : north zone
2 : south zone
3 : east zone
4 : west zone
Enter first junction ID: 0
Enter second junction ID: 2
Enter road construction cost: 3
Road added successfully!

===== SMART CITY ROAD NETWORK =====
1. Add Junction
2. Add Road and Construction Cost
3. Display Junctions
4. Display Available Roads
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 2

--- Smart City Junctions ---
0 : city centre
1 : north zone
2 : south zone
3 : east zone
4 : west zone
Enter first junction ID: 1
Enter second junction ID: 2
Enter road construction cost: 2
Road added successfully!

===== SMART CITY ROAD NETWORK =====
1. Add Junction
2. Add Road and Construction Cost
3. Display Junctions
4. Display Available Roads
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 2

--- Smart City Junctions ---
0 : city centre
1 : north zone
2 : south zone
```

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--- Smart City Junctions ---
0 : city centre
1 : north zone
2 : south zone
3 : east zone
4 : west zone
Enter first junction ID: 1
Enter second junction ID: 3
Enter road construction cost: 5
Road added successfully!

===== SMART CITY ROAD NETWORK =====
1. Add Junction
2. Add Road and Construction Cost
3. Display Junctions
4. Display Available Roads
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 2

--- Smart City Junctions ---
0 : city centre
1 : north zone
2 : south zone
3 : east zone
4 : west zone
Enter first junction ID: 2
Enter second junction ID: 3
Enter road construction cost: 3
Road added successfully!

===== SMART CITY ROAD NETWORK =====
1. Add Junction
2. Add Road and Construction Cost
3. Display Junctions
4. Display Available Roads
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 3

--- Smart City Junctions ---
0 : city centre
1 : north zone
2 : south zone
```

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```
--- Smart City Junctions ---
0 : city centre
1 : north zone
2 : south zone
3 : east zone
4 : west zone

===== SMART CITY ROAD NETWORK =====
1. Add Junction
2. Add Road and Construction Cost
3. Display Junctions
4. Display Available Roads
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 4

--- Available Roads ---
From Junction           To Junction             Cost
city centre             north zone              4
city centre             south zone              3
north zone              south zone              2
north zone              east zone               5
south zone              east zone               3

===== SMART CITY ROAD NETWORK =====
1. Add Junction
2. Add Road and Construction Cost
3. Display Junctions
4. Display Available Roads
5. Find Minimum-Cost Network (Kruskal)
6. Exit
Enter your choice: 5

--- Minimum-Cost Smart City Road Network ---
From Junction           To Junction             Cost
north zone              south zone              2
city centre             south zone              3
south zone              east zone               3

The road network is disconnected.
Not all junctions can be connected with the roads entered.
Total minimum construction cost: 8
Roads selected: 3
```